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Huang et al.

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(54) **INTEGRATED CIRCUIT DEVICE AND METHOD**

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H10D 84/90 (2025.01)
G06F 30/392 (2020.01)
G06F 119/06 (2020.01)
H10D 84/85 (2025.01)
H10D 89/10 (2025.01)

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CPC **H10D 84/907** (2025.01); **G06F 30/392** (2020.01); **H10D 84/854** (2025.01); **H10D 89/10** (2025.01); **G06F 2119/06** (2020.01); **H10D 84/853** (2025.01); **H10D 84/961** (2025.01); **H10D 84/991** (2025.01)

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CPC **H10D 84/961**; **H10D 84/991**; **H10D 89/10**; **H01L 2027/11861**; **H01L 2027/1189**; **H01L 27/0207**

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

7,260,442 B2 8/2007 Hwang et al.
9,256,709 B2 2/2016 Yu et al.
(Continued)

FOREIGN PATENT DOCUMENTS

CN 108133933 6/2018
CN 109599400 4/2019
CN 110797337 2/2020

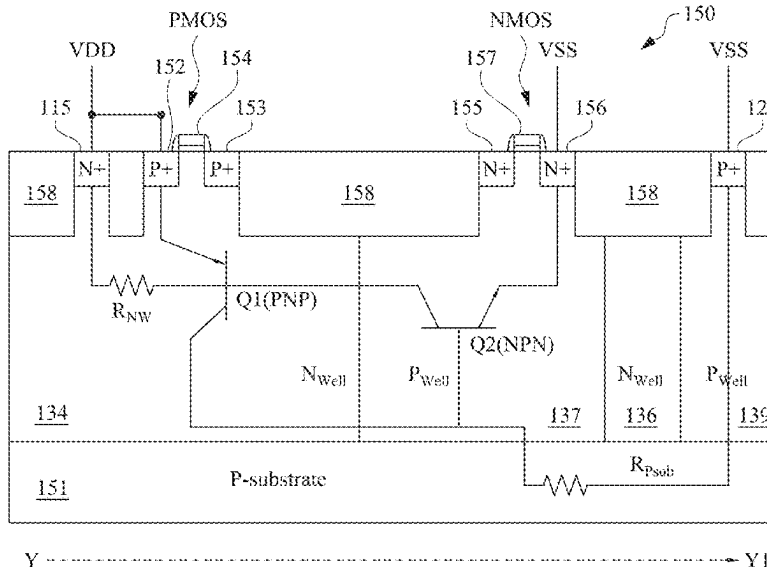
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(57) **ABSTRACT**

An integrated circuit (IC) device includes a plurality of first doped regions of a first semiconductor type over at least one first well region of the first semiconductor type, and a second doped region of a second semiconductor type over a second well region of the second semiconductor type. The second semiconductor type is different from the first semiconductor type. The plurality of first doped regions is arranged along a first direction. Each of the plurality of first doped regions has a first length in the first direction. The second doped region extends in the first direction between at least two first doped regions among the plurality of first doped regions over a second length greater than the first length.

20 Claims, 12 Drawing Sheets



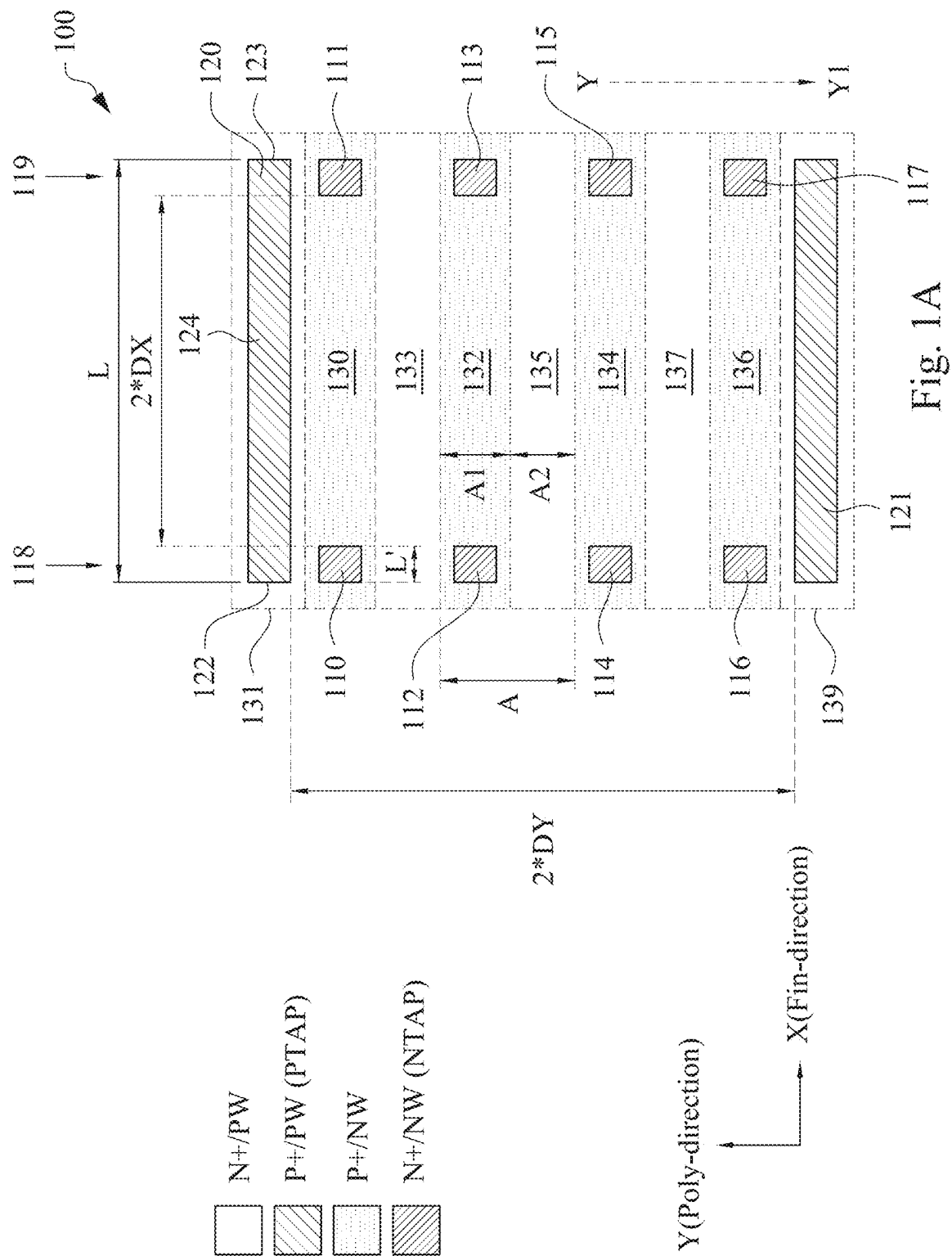
(56)

References Cited

U.S. PATENT DOCUMENTS

2003/0218841	A1 *	11/2003	Kodama		H01L 27/0251
					361/56
2008/0094869	A1	4/2008	Osada et al.		
2012/0168875	A1	7/2012	Tamaru et al.		
2014/0040838	A1	2/2014	Liu et al.		
2015/0278429	A1	10/2015	Chang		
2018/0137231	A1	5/2018	Sakuda		
2019/0065659	A1	2/2019	Biswas et al.		

* cited by examiner



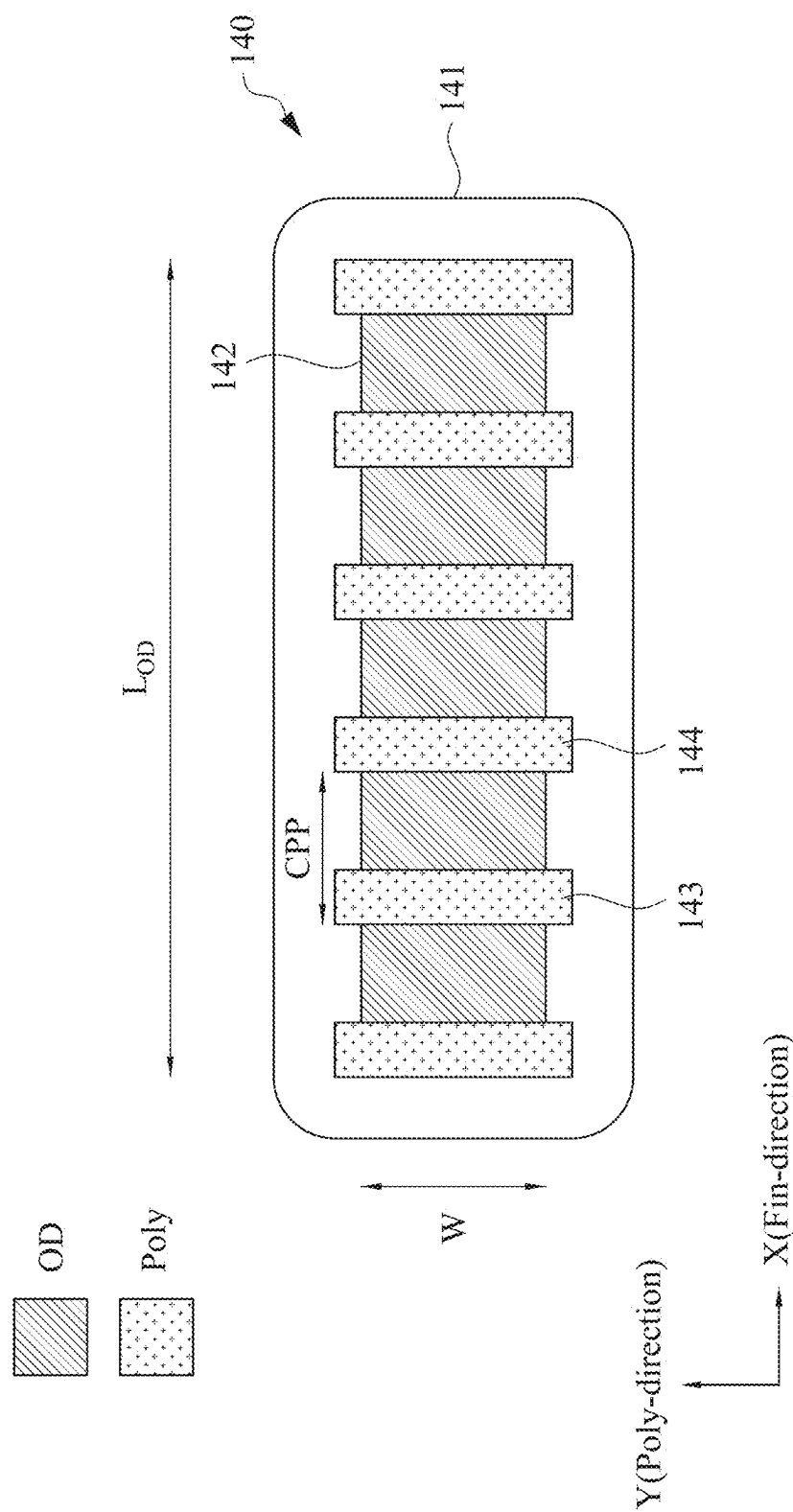
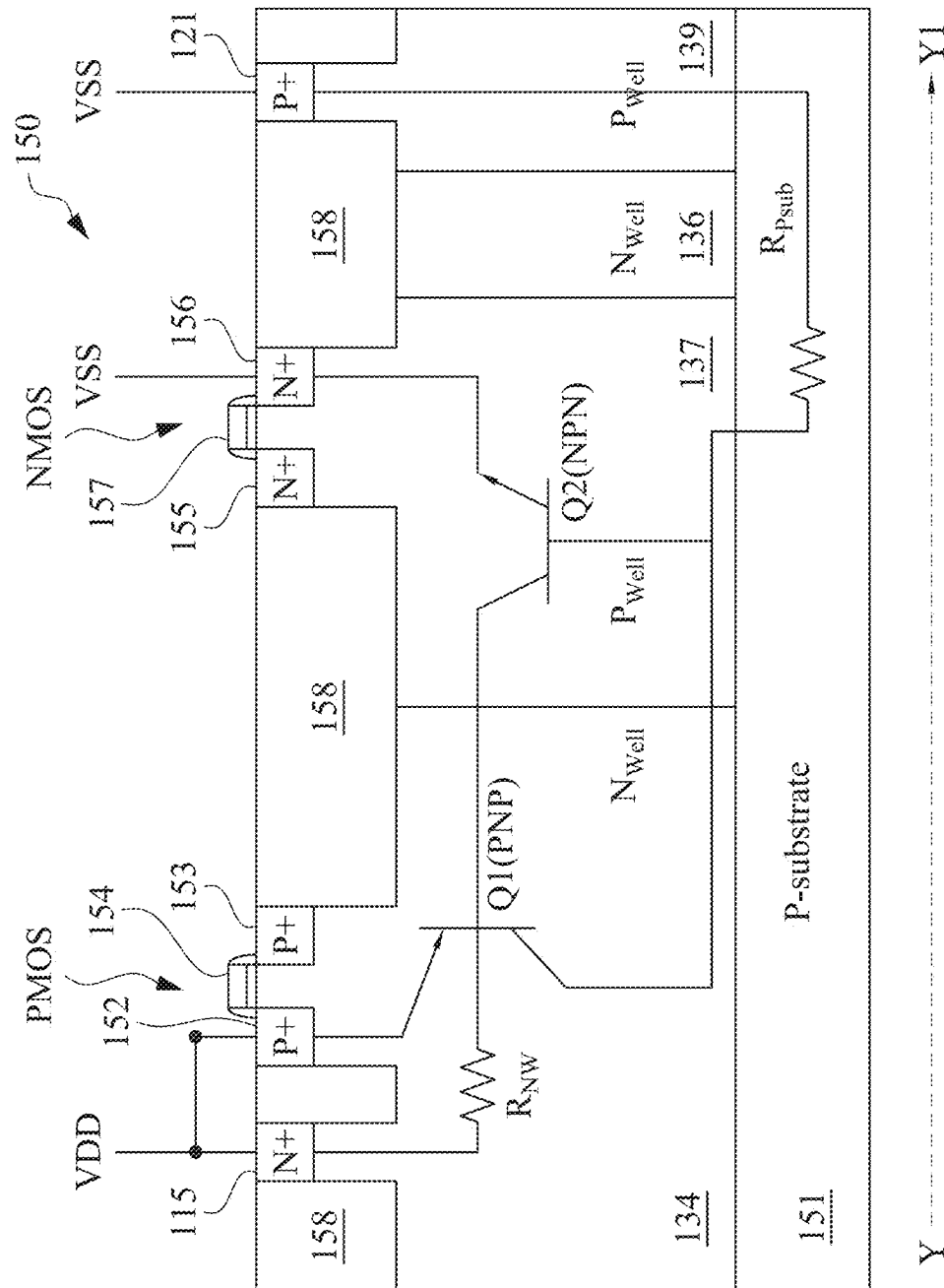


Fig. 1B



100

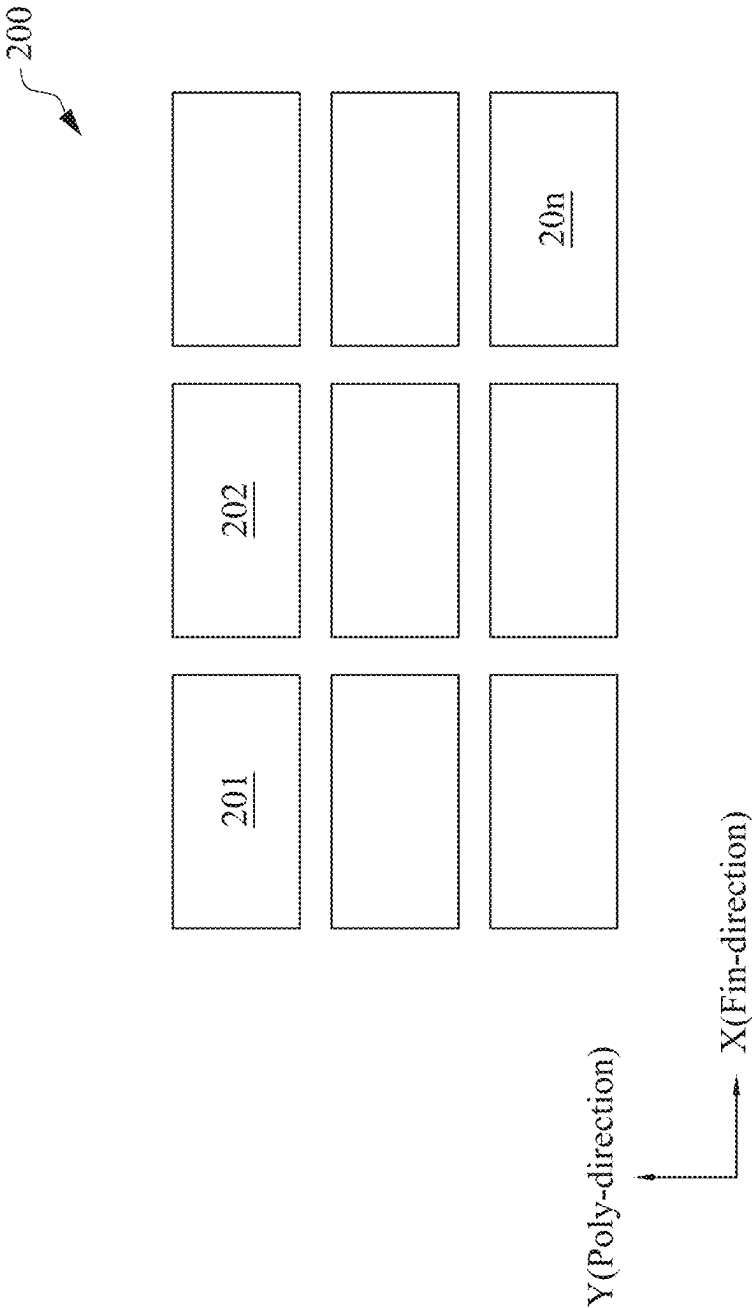
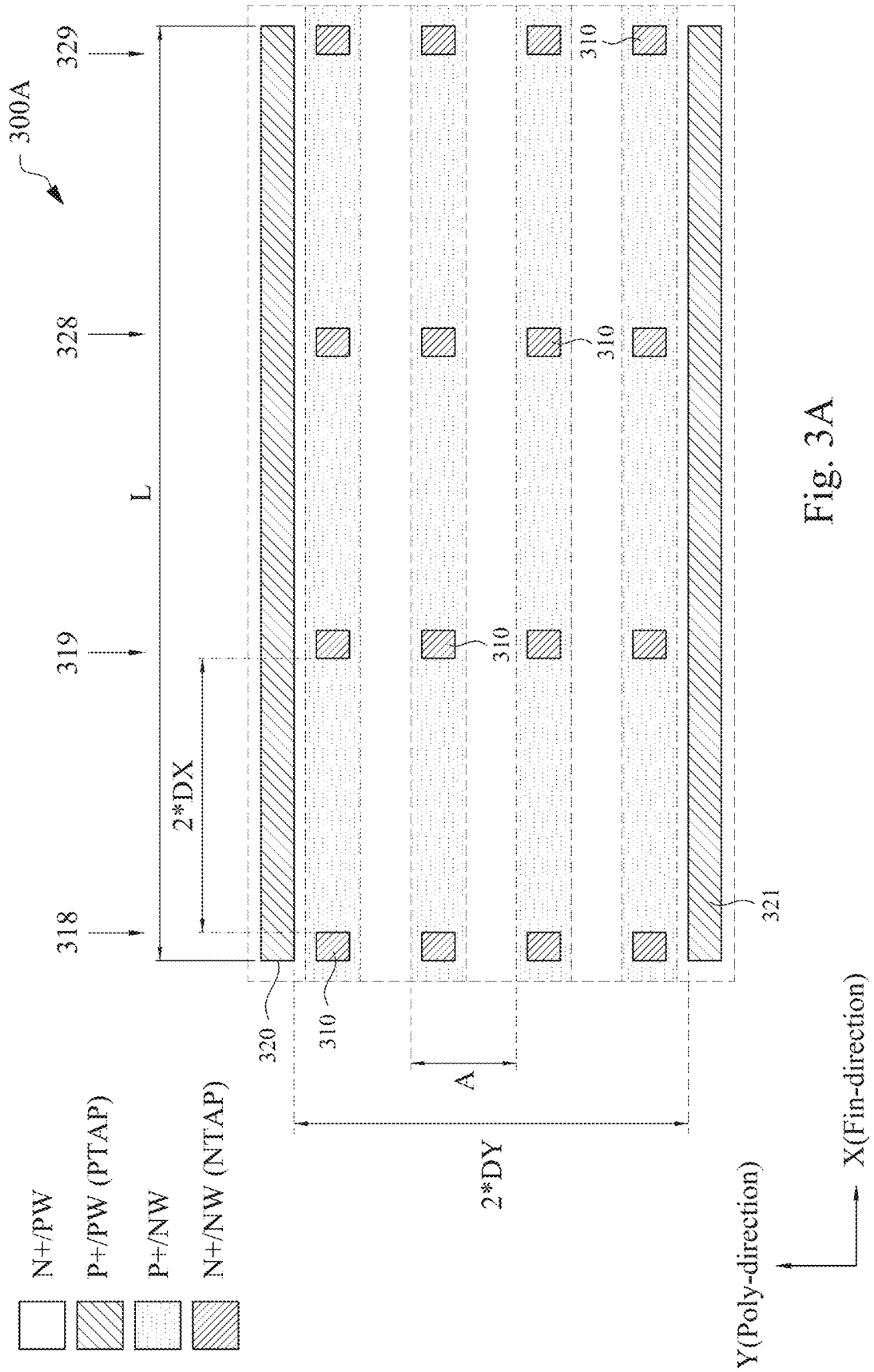
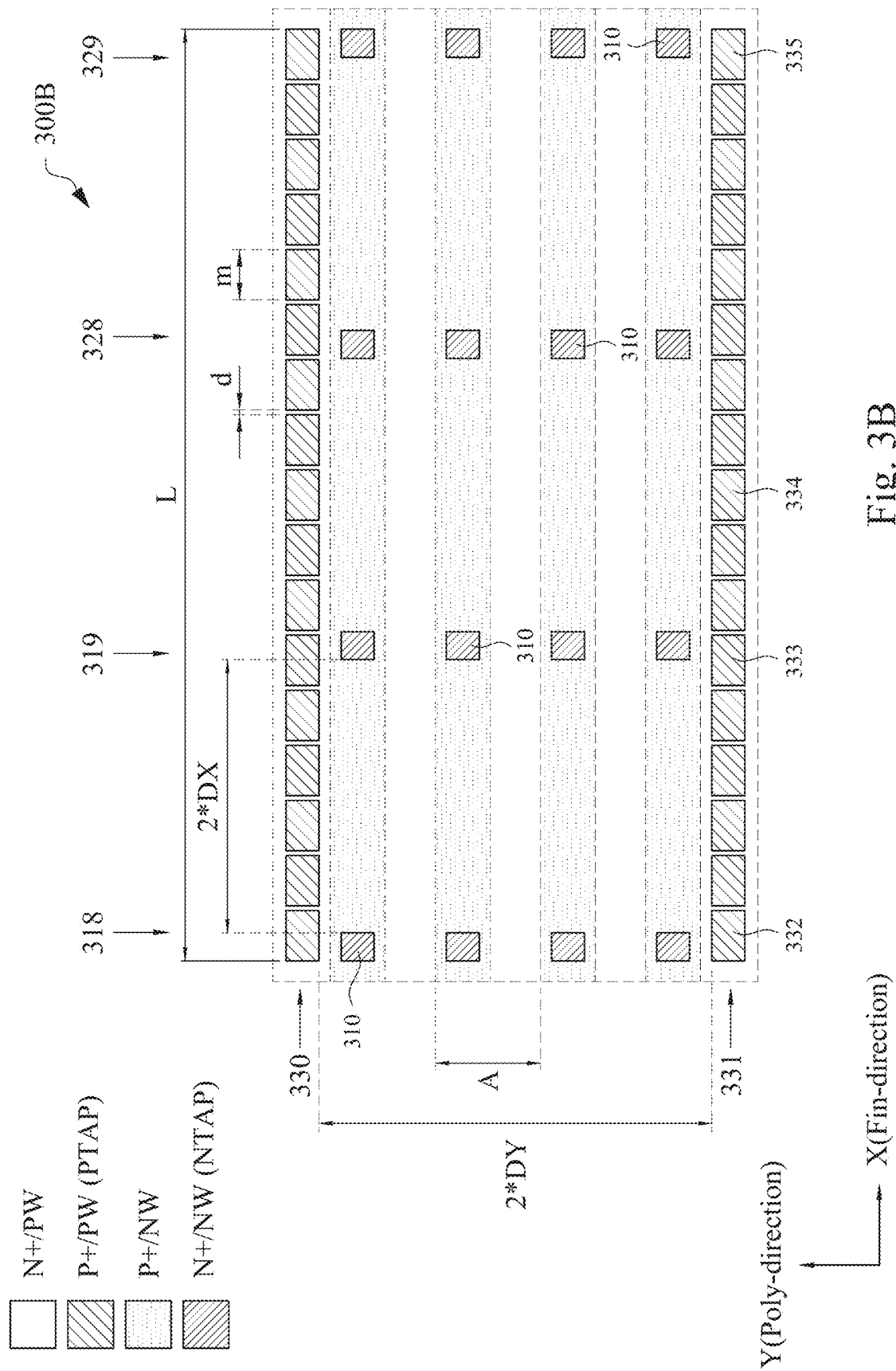
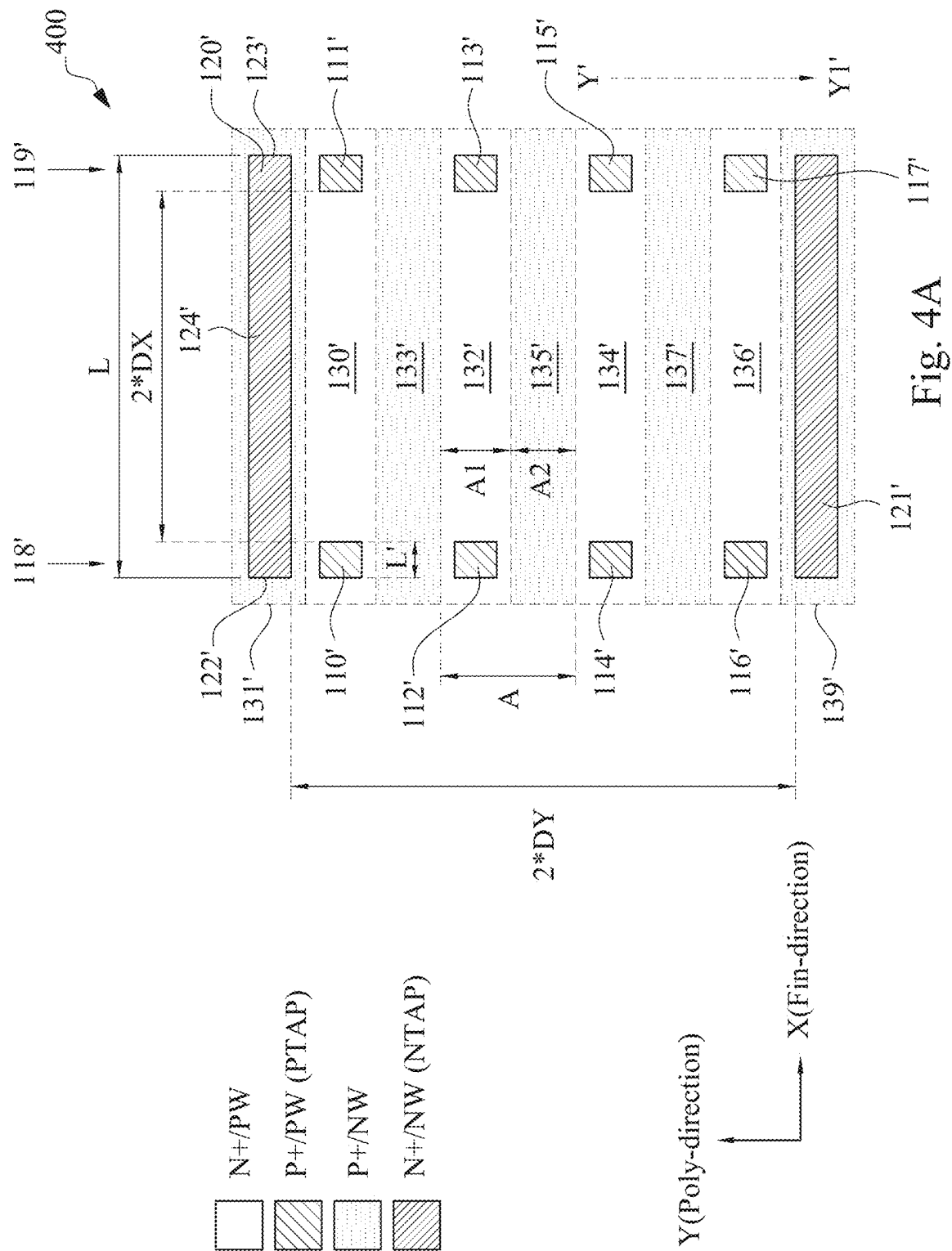
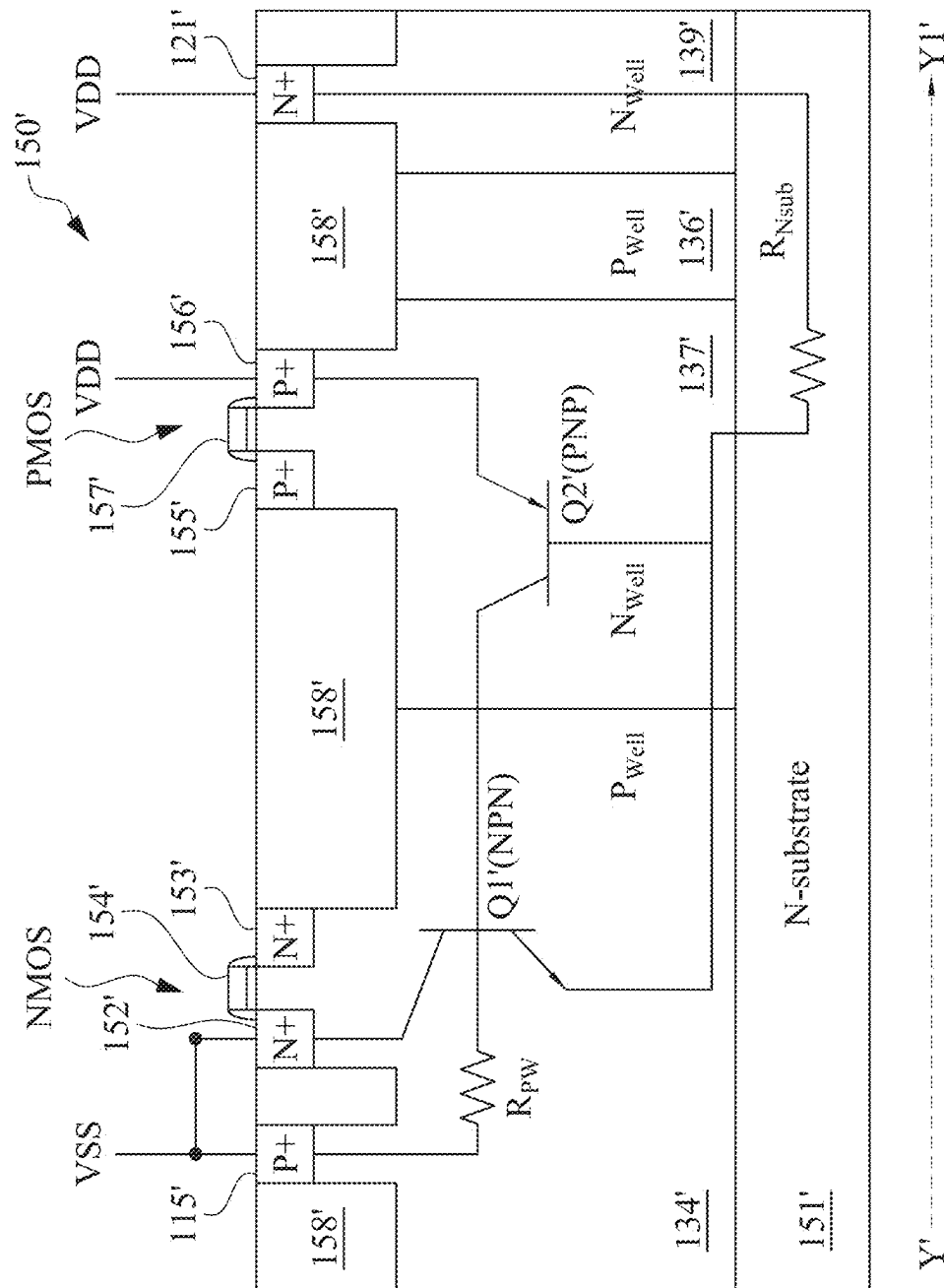


Fig. 2









34
b6
b7C
b7D

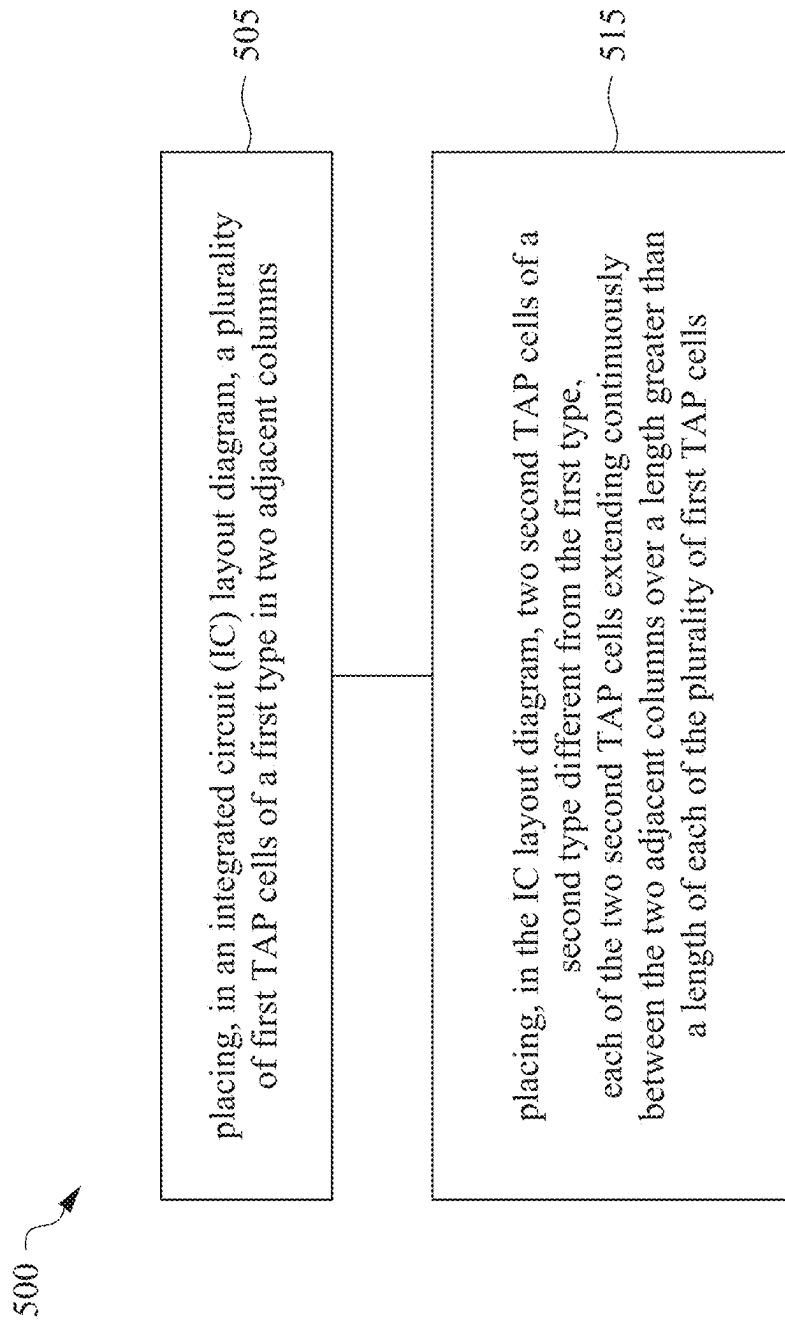


Fig. 5

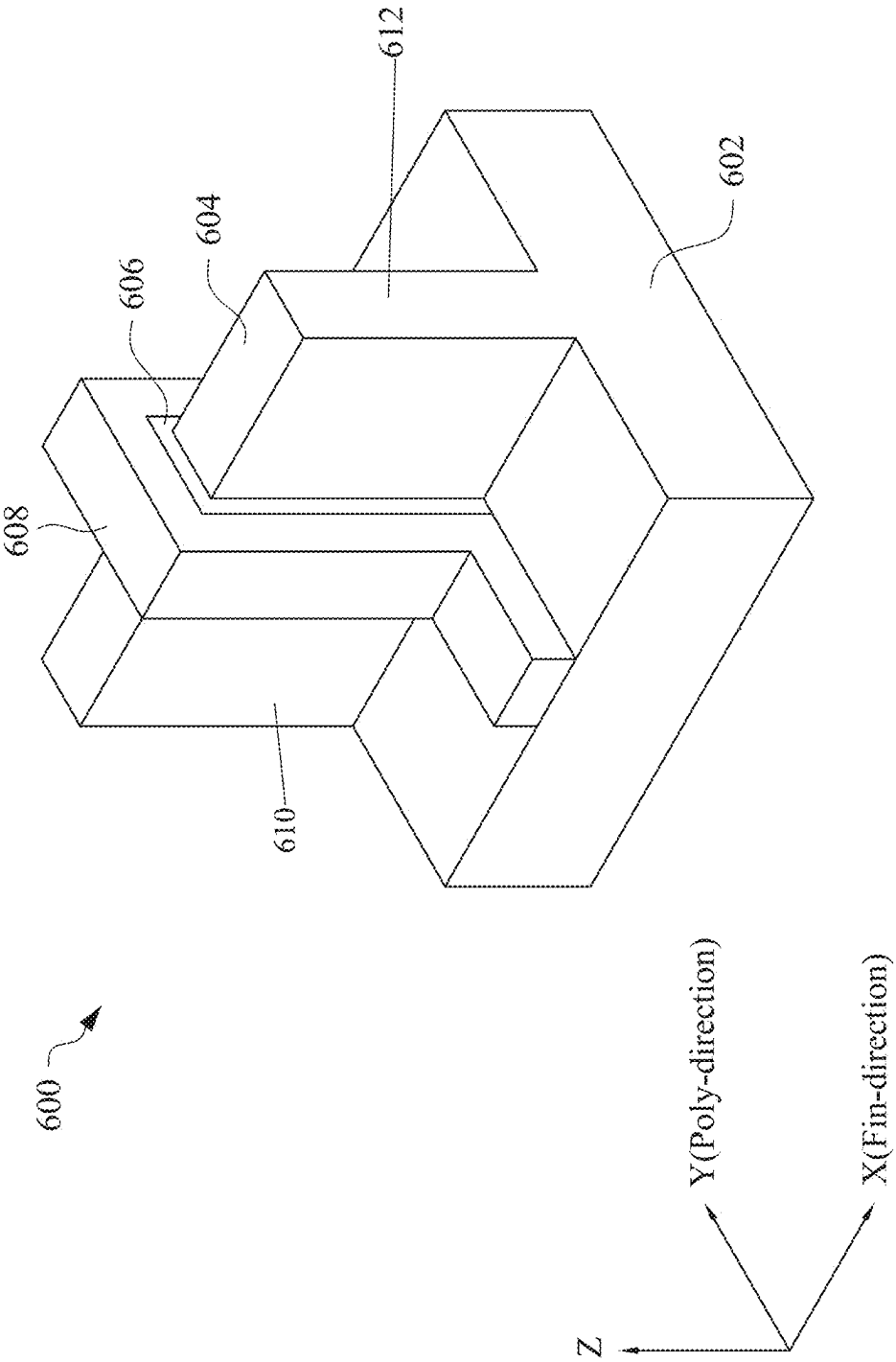


Fig. 6

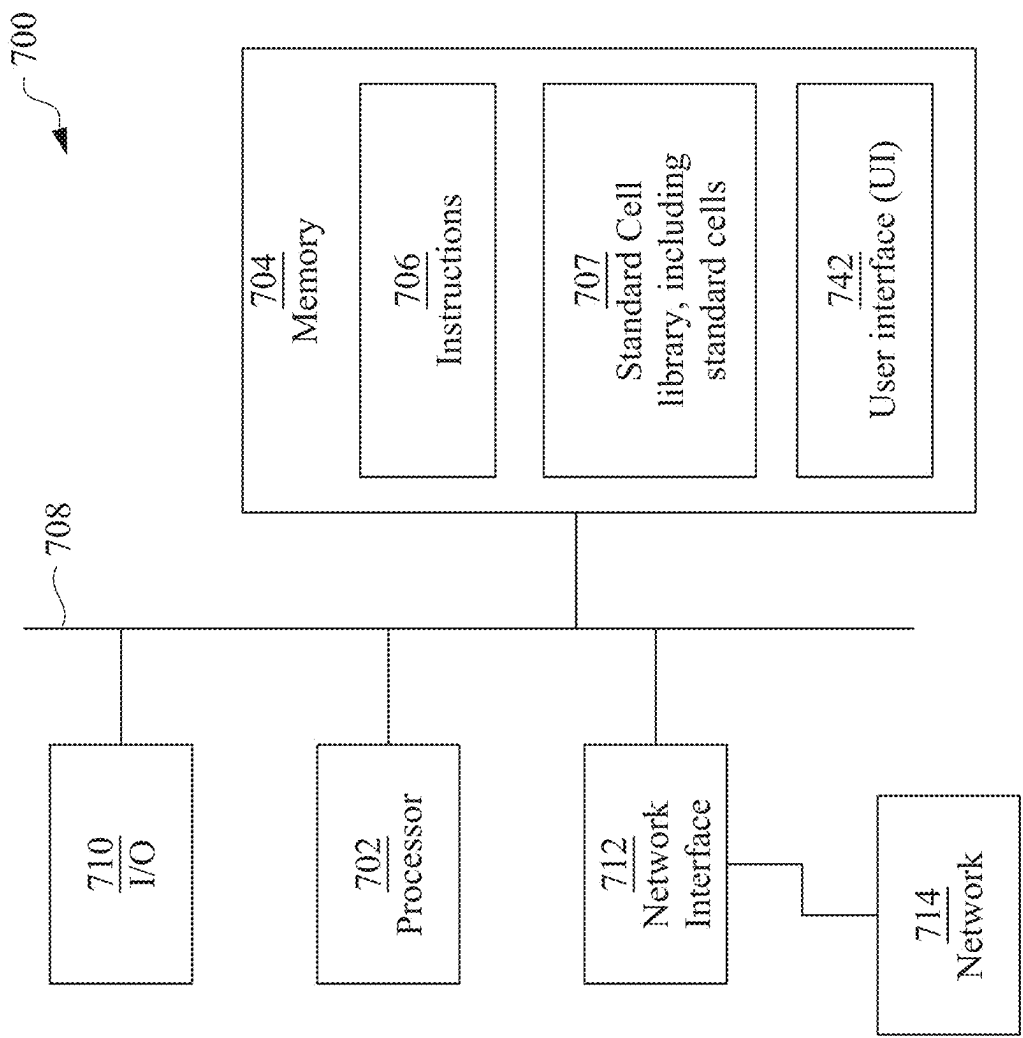


Fig. 7

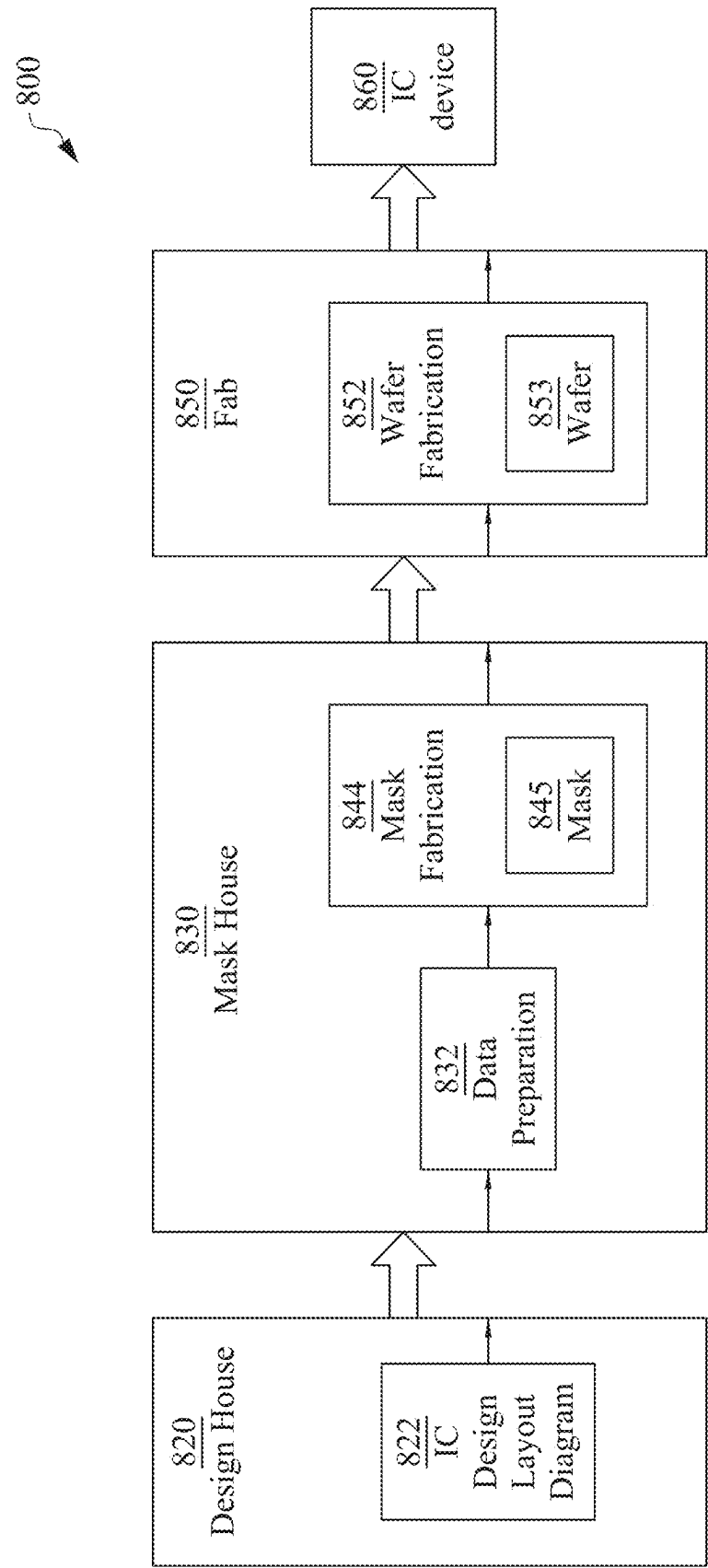


Fig. 8

INTEGRATED CIRCUIT DEVICE AND METHOD

PRIORITY CLAIM

The present application is a continuation application of U.S. application Ser. No. 17/024,351, filed Sep. 17, 2020, now U.S. Pat. No. 11,646,317, issued May 9, 2023, which claims the priority of U.S. Provisional Application No. 62/982,488, filed Feb. 27, 2020. The above-referenced patent(s) and applications are incorporated herein by reference in their entireties.

BACKGROUND

An integrated circuit (IC) typically includes a number of semiconductor devices represented in an IC layout diagram. An IC layout diagram is hierarchical and includes modules which carry out higher-level functions in accordance with the semiconductor device's design specifications. The modules are often built from a combination of cells, each of which represents one or more semiconductor structures configured to perform a specific function. Cells having pre-designed layout diagrams, sometimes known as standard cells, are stored in standard cell libraries (hereinafter "libraries" or "cell libraries" for simplicity) and accessible by various tools, such as electronic design automation (EDA) tools, to generate, optimize and verify designs for ICs.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1A is a schematic view of an IC layout diagram, in accordance with some embodiments.

FIG. 1B is a schematic, enlarged view of a portion of an IC layout diagram, in accordance with some embodiments.

FIG. 1C is a schematic cross-sectional view combined with a schematic circuit diagram of an IC device, in accordance with some embodiments.

FIG. 2 is a schematic view of an IC layout diagram, in accordance with some embodiments.

FIG. 3A is a schematic view of an IC layout diagram, in accordance with some embodiments.

FIG. 3B is a schematic view of an IC layout diagram, in accordance with some embodiments.

FIG. 4A is a schematic view of an IC layout diagram, in accordance with some embodiments.

FIG. 4B is a schematic cross-sectional view combined with a schematic circuit diagram of an IC device, in accordance with some embodiments.

FIG. 5 is a flow chart of a method of generating an IC layout diagram, in accordance with some embodiments.

FIG. 6 is a perspective view of an example transistor having a fin feature, in accordance with some embodiments.

FIG. 7 is a block diagram of an EDA system, in accordance with some embodiments.

FIG. 8 is a block diagram of an IC manufacturing system and an IC manufacturing flow associated therewith, in accordance with some embodiments.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different fea-

tures of the provided subject matter. Specific examples of components, materials, values, steps, operations, materials, arrangements, or the like, are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. Other components, values, operations, materials, arrangements, or the like, are contemplated. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed. Source/drain(s) may refer to a source or a drain, individually or collectively dependent upon the context.

Further, spatially relative terms, such as "beneath," "below," "lower," "above," "upper" and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

A well tap cell, referred to herein as "TAP cell," is a standard cell which defines a region in a doped well where the doped well is coupled to a bias voltage, such as a power supply voltage. TAP cells are included in an IC layout diagram to improve latch-up immunity of ICs manufactured in accordance with the IC layout diagram.

With the current tendency of scaling down semiconductor devices, placement of TAP cells in an IC layout diagram for manufacturing ICs raises one or more considerations including, but not limited to, process bottleneck due to reduced lithography critical dimension (CD), and mixed channel effects. To address one or more of such considerations, in an IC layout diagram in accordance with some embodiments, first TAP cells of a first semiconductor type (e.g., N-type or P-type) are placed in rows and columns, and second TAP cells of a different, second semiconductor type (e.g., P-type or N-type) are placed in elongated configurations or band shapes across multiple columns of the first TAP cells. As a result, in at least one embodiment, it is possible to achieve one or more effects, including, but not limited to, relaxing of process constraints, increasing of latch-up immunity, reducing of areas occupied or blocked by TAP cells, and increasing of areas where standard cells other than TAP cells are placeable.

FIG. 1A is a schematic view of an IC layout diagram 100 of an IC device, in accordance with some embodiments. The IC layout diagram 100 comprises a plurality of first TAP cells 110-117 of a first semiconductor type, and a plurality of second TAP cells 120, 121 of a second semiconductor type different from the first semiconductor type. The first TAP cells 110-117 are arranged in at least two columns 118, 119. For example, the first TAP cells 110, 112, 114, 116 are arranged in the column 118, and the first TAP cells 111, 113, 115, 117 are arranged in the column 119. The columns 118, 119 are adjacent each other in a first direction, e.g., X direction, and extend in a second direction, e.g., Y direction,

transverse to the first direction. At least one of the second TAP cells **120**, **121** extends in the X direction between the columns **118**, **119** over a length L greater than a length L' of each of the first TAP cells **110-117** in the X direction. The at least one of the second TAP cells **120**, **121** overlaps, in the Y direction along the page, at least one first TAP cell in at least one of the columns **118**, **119**.

In the example configuration in FIG. 1A, the second TAP cell **120** is elongated in the X direction, and has a length L in the X direction greater than its height in the Y direction. The second TAP cell **120** overlaps, in the Y direction, the first TAP cells **110-117**. For example, the second TAP cell **120** extends continuously in the X direction from a first end **122** to a second end **123** thereof. The first end **122** of the second TAP cell **120** overlaps, in the Y direction, the first TAP cells **110**, **112**, **114**, **116** in the column **118**. The second end **123** of the second TAP cell **120** overlaps, in the Y direction, the first TAP cells **111**, **113**, **115**, **117** in the column **119**. A middle portion **124** of the second TAP cell **120** is between the first end **122** and the second end **123**, and overlaps none of the first TAP cells **110-117** in the Y direction. The second TAP cell **121** has an elongated configuration similar to that described above with respect to the second TAP cell **120**. The second TAP cells **120**, **121** are adjacent each other in the Y direction, and there is no other TAP cell of the second semiconductor type between the second TAP cells **120**, **121**. The second TAP cells **120**, **121** sandwich therebetween multiple rows and columns of the first TAP cells, namely, the two columns **118**, **119**, and four rows formed respectively by the first TAP cells **110** and **111**, the first TAP cells **112** and **113**, the first TAP cells **114** and **115**, and the first TAP cells **116** and **117**.

The configuration described above with respect to FIG. 1A is an example, and other configurations are within the scopes of various embodiments. For example, some embodiments include different numbers of first TAP cells in each of the columns **118**, **119** and/or different numbers of columns of first TAP cells sandwiched between each pair of adjacent second TAP cells **120**, **121**. In at least one embodiment, at least one of the second TAP cells **120**, **121** has a configuration different from that described above with respect to FIG. 1A. For example, at least one of the first end **122** or second end **123** of the second TAP cell **120** does not overlap the corresponding column **118** or **119** in the Y direction. For another example, the second TAP cells **120**, **121** have different lengths in the X direction. In a further example, at least one of the second TAP cells **120**, **121** is not a single second TAP cell extending continuously in the X direction as described with respect to FIG. 1A, but instead comprises a series of discrete second TAP cells as described herein with respect to FIG. 3B.

The IC layout diagram **100** further comprises a plurality of first well regions **130**, **132**, **134**, **136** of the first semiconductor type, and a plurality of second well regions **131**, **133**, **135**, **137**, **139** of the second semiconductor type. The first well regions **130**, **132**, **134**, **136** and the second well regions **131**, **133**, **135**, **137**, **139** extend in the X direction, and are arranged alternately in the Y direction. Each of the first TAP cells **110-117** is in a corresponding one of the first well regions **130**, **132**, **134**, **136**, and each of the second TAP cells **120**, **121** is in a corresponding one of the second well regions **131**, **133**, **135**, **137**, **139**. For example, the first TAP cells **110** and **111** are in the first TAP cell **130**, the first TAP cells **112** and **113** are in the first TAP cell **132**, the first TAP cells **114** and **115** are in the first TAP cell **134**, and the first TAP cells **116** and **117** are in the first TAP cell **136**, whereas

the second TAP cell **120** is in the second well region **131**, and the second TAP cell **121** is in the second well region **139**.

In the example configuration in FIG. 1A, the first semiconductor type is N-type and the second semiconductor type is P-type. In other words, the first well regions **130**, **132**, **134**, **136** are N-type well regions (hereinafter "N wells"), the second well regions **131**, **133**, **135**, **137**, **139** are P-type well regions (hereinafter "P wells"), the first TAP cells **110-117** are N-type TAP cells (hereinafter "NTAP cells"), and the second TAP cells **120**, **121** are P-type TAP cells (hereinafter "PTAP cells"). An N well is a region that includes N-type dopants, whereas a P well is a region that includes P-type dopants. In the drawings, N wells are labelled as "NW," P wells are labelled as "PW," N-type dopants are labelled as "N+" and P-type dopants are labelled as "P+."

An NTAP cell is a region in an N well, but with a higher concentration of N-type dopants than the N well itself. For example, the NTAP cell **110** has a higher concentration of N-type dopants than the N well **130** in which the NTAP cell **110** is formed. A PTAP cell is a region in a P well, but with a higher concentration of P-type dopants than the P well itself. For example, the PTAP cell **120** has a higher concentration of P-type dopants than the P well **131** in which the PTAP cell **120** is formed.

In an N well, P-type active regions with P-type dopants are arranged in areas not occupied or blocked by NTAP cells to form one or more circuit elements. In a P well, N-type active regions with N-type dopants are arranged in areas not occupied or blocked by PTAP cells to form one or more circuit elements. Examples of circuit elements include, but are not limited to, transistors and diodes. Examples of transistors include, but are not limited to, metal oxide semiconductor field effect transistors (MOSFET), complementary metal oxide semiconductor (CMOS) transistors, bipolar junction transistors (BJT), high voltage transistors, high frequency transistors, P-channel and/or N-channel field effect transistors (PFETs/NFETs), etc.), FinFETs, planar MOS transistors with raised source/drains, or the like. For example, in the N wells **130**, **132**, **134**, **136**, P-type active regions (not shown in FIG. 1A, but described herein with respect to FIG. 1C) are arranged in the areas not occupied or blocked by the NTAP cells **110-117** to define P-channel metal-oxide semiconductor (PMOS) regions for forming PMOS transistors. In the P wells **133**, **135**, **137**, N-type active regions (not shown in FIG. 1A, but described herein with respect to FIG. 1C) are arranged in the areas not occupied or blocked by the PTAP cells **120**, **121** to define N-channel metal-oxide semiconductor (NMOS) regions for forming NMOS transistors. A cell having a pre-designed layout diagram is read from a cell library and placed in the IC layout diagram **100** such that NMOS transistors of the cell are arranged in an NMOS region, whereas PMOS transistors of the cell are arranged in a PMOS region. NTAP cells, PTAP cells, N-type active regions and P-type active regions are sometimes commonly referred to as oxide-definition (OD) regions, and are schematically illustrated in FIG. 1B with the label "OD."

The IC layout diagram **100** further comprises gate regions (not shown in FIG. 1A, but described herein with respect to FIG. 1B) which include a conductive material, such as, polysilicon, and are schematically illustrated in FIG. 1B with the label "Poly." Other conductive materials for the gate regions, such as metals, are within the scopes of various embodiments. The gate regions extend, or are elongated, in the Y direction across the OD regions. The Y direction is also referred to herein as the Poly direction. In some embodi-

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ments, each OD region has one or more fin features arranged therein. Such fin features extend, or are elongated, in the X direction, and spaced from each other in the Y direction. The X direction is also referred to herein as the Fin direction. An example of a fin feature is described with respect to FIG. 6.

In at least one embodiment, a number of fin features in each of the first TAP cells **110-117** and a number of fin features in each of the second TAP cells **120, 121** satisfy the following relationship:

$$F2/F1 \geq (DY/DX) * (L/A) \quad (1)$$

where

F1 is the number of fin features in each of the first TAP cells **110-117**,

F2 is the number of fin features in each of second TAP cells **120, 121** which are elongated in the X direction,

DX is a half of a first distance $2 * DX$ in the X direction between facing sides of the first TAP cells in the adjacent columns **118, 119**,

DY is a half of a second distance $2 * DY$ in the Y direction between facing sides of the second TAP cells **120, 121**,

L is the length of the second TAP cell **120** or **121** in the X direction, and

A is a cell height in the Y direction, and is a sum of a height A1 of a first well region (e.g., **132**) and a height A2 of an adjacent second well region (e.g., **135**).

In the example configuration in FIG. 1A, F1 is the number F_{NTAP} of fin features in each NTAP cell, and F2 is the number F_{PTAP} of fin features in each PTAP cell, and the relationship (1) becomes

$$F_{PTAP}/F_{NTAP} \geq (DY/DX) * (L/A) \quad (1')$$

By configuring the NTAP cells and PTAP cells to have different configurations and to satisfy the relationship (1) or (1'), it is possible in some embodiments to match or improve a latch-up (LUP) immunity index compared to another approach. Specifically, the LUP immunity index of an IC device corresponding to the IC layout diagram **100** is determined by the following relationship

$$V = J_{body} * L * (A/2) * (2DY/A) * (Rc/F_{PTAP}) \quad (2)$$

where

V is the LUP immunity index represented by a voltage drop caused by a leakage current J_{body} in the IC device, and Rc is a unit resistance.

The lower the voltage drop V, the better LUP immunity of the IC device.

In another approach where the NTAP cells and PTAP cells are configured similarly to each other, and similarly to the NTAP cells **110-117**, the LUP immunity index V' of an IC device in accordance with the another approach is determined by the following relationship

$$V' = J_{body} * 2DX * (A/2) * (Rc/F_{NTAP}) \quad (3)$$

To match or improve the LUP immunity index compared to the another approach, the following relationship is to be satisfied:

$$V \leq V' \quad (4)$$

Based on the relationships (2), (3) and (4), the relationships (1') and (1) are obtained.

In some embodiments, the IC layout diagram **100** satisfies at least one of the following: DY is from $0.5 \mu\text{m}$ to $1000 \mu\text{m}$, DX is from $0.05 \mu\text{m}$ to $100 \mu\text{m}$, L is from $0.1 \mu\text{m}$ to $5000 \mu\text{m}$, or A is from $0.025 \mu\text{m}$ to $0.300 \mu\text{m}$. The range of $0.025 \mu\text{m}$ to $0.300 \mu\text{m}$ of the cell height A corresponds to one or more considerations and/or constraints in an example semiconductor manufacturing process. At the range of $0.025 \mu\text{m}$

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to $0.300 \mu\text{m}$ of the cell height A, if DX is below the range of $0.05 \mu\text{m}$ to $100 \mu\text{m}$ and/or if DY is below the range of $0.5 \mu\text{m}$ to $1000 \mu\text{m}$, there is an excessive increase in the chip area of TAP cells with an associated decrease in the remaining chip area for other functional cells of the IC layout diagram **100**. At the range of $0.025 \mu\text{m}$ to $0.300 \mu\text{m}$ of the cell height A, if DX is above the range of $0.05 \mu\text{m}$ to $100 \mu\text{m}$ and/or if DY is above the range of $0.5 \mu\text{m}$ to $1000 \mu\text{m}$, there is an elevated risk of latch-up. The range of $0.1 \mu\text{m}$ to $5000 \mu\text{m}$ for the length L of the elongated TAP cells (e.g., **120** or **121**) is derived from the respective range(s) for A, DX and/or DY based on the relationship (1) or (1').

FIG. 1B is a schematic, enlarged view of a portion **140** of the IC layout diagram **100**, in accordance with some embodiments, for describing an example to determine the number of fin features in a TAP cell. The portion **140** includes a well region **141**, an OD region **142**, and a plurality of gate regions **143, 144**. The well region **141** extends in the X direction, and surrounds or encloses therein the OD region **142**. The OD region **142** includes one or more fin features (not shown) which extend in the X direction. The OD region **142** has a length L_{OD} in the X direction, and a height W in the Y direction. The gate regions **143, 144** extend in the Y direction across the OD region **142**, and are arranged at a pitch CPP in the X direction.

The number F of fin features in the OD region **142** is determined by

$$F = (L_{OD}/CPP) * W * F_n \quad (5)$$

where F_n is a number of fin features per unit height in the Y direction. In at least one embodiment, CPP and W are the same for all TAP cells.

In an example, the portion **140** corresponds to a region containing a first TAP cell, e.g., **110** in FIG. 1A. Specifically, the well region **141** corresponds to the first well region **130**, the OD region **142** corresponds to the first TAP cell **110**, L_{OD} corresponds to the length L' of the first TAP cell **110**, W corresponds to the height of the first TAP cell **110** in the Y direction, and F corresponds to F1 or F_{NTAP} which is the number of fin features in the first TAP cell **110**. Accordingly, the number of fin features in each first TAP cell is determinable from the relationship (5).

In another example, the portion **140** corresponds to a region containing a second TAP cell, e.g., **120** in FIG. 1A. Specifically, the well region **141** corresponds to the second well region **131**, the OD region **142** corresponds to the second TAP cell **120**, L_{OD} corresponds to the length L of the second TAP cell **120**, W corresponds to the height of the second TAP cell **120** in the Y direction, and F corresponds to F2 or F_{PTAP} which is the number of fin features in the second TAP cell **120**. Accordingly, the number of fin features in each second TAP cell is also determinable from the relationship (5).

FIG. 1C is a schematic cross-sectional view of an IC device **150**, in accordance with some embodiments. The IC device **150** corresponds to a portion of the IC layout diagram **100** which is indicated by arrow $Y \rightarrow Y1$ in FIG. 1A. The cross-sectional view in FIG. 1C is also combined with a schematic circuit diagram of the IC device **150**.

The IC device **150** comprises a substrate **151** on which the TAP cells, well regions, active regions, gate regions, fin features described with respect to FIGS. 1A and 1B are formed. For example, the IC device **150** comprises, on the substrate **151**, the N well **134**, P well **137**, N well **136**, and P well **139**. P-type active regions **152, 153**, and the NTAP cell **115** are formed in the N well **134**. A gate region **154** is formed over the P-type active regions **152, 153**, and defines

together with the P-type active regions **152**, **153** a PMOS. N-type active regions **155**, **156** are formed in the P well **137**. A gate region **157** is formed over the N-type active regions **155**, **156**, and defines together with the N-type active regions **155**, **156** an NMOS. The PTAP cell **121** is formed in the P well **139**. The IC device **150** further comprises a plurality of isolation regions **158** between adjacent P well and N well. The P-type active region **152** of the PMOS is coupled to a first power supply voltage, e.g., VDD. The N-type active region **156** of the NMOS is coupled to a second power supply voltage, e.g., VSS, which is in at least one embodiment, the ground. The substrate **151** is a P-type substrate.

The schematic circuit diagram of the IC device **150** in FIG. **1C** shows parasitic transistors **Q1** and **Q2**. The parasitic transistor **Q1** is a PNP transistor formed by the P-type active region **152**, the N well **134**, and the P-type substrate **151**. The parasitic transistor **Q2** is an NPN transistor formed by the N well **134**, the P well **137**, and the N-type active region **156**. In the absence of the NTAP cell **115** and/or the PTAP cell **121**, there is a concern that a leakage current in one or more of the P-type substrate **151**, P wells and N wells of the IC device **150** is sufficient to cause both of the parasitic transistors **Q1** and **Q2** to turn ON, and create a current path from VDD, through the turned ON parasitic transistors **Q1** and **Q2**, to VSS. Such a current path between VDD and VSS is a latch-up situation that adversely affects performance of the IC device **150**.

The provision of the NTAP cell **115** which is coupled to VDD and the PTAP cell **121** which is coupled to VSS reduces the likelihood of latch-up situations and improves LUP immunity of the IC device **150**. In the schematic circuit diagram of the IC device **150** in FIG. **1A**, a resistor R_{NW} represents a TAP cell resistance between the NTAP cells, representative by the NTAP cell **115**, and the base of the parasitic transistor **Q1**, whereas a resistor R_{Psub} represents a TAP cell resistance between the PTAP cells, representative by the PTAP cell **121**, and the base of the parasitic transistor **Q2**. The lower the resistances of the resistors R_{NW} and R_{Psub} , the lower the likelihood of the parasitic transistors **Q1** and **Q2** being turned ON, respectively, the better the LUP immunity of the IC device **150**. The resistance of the resistor R_{NW} depends on a configuration and/or arrangement of NTAP cells. The resistance of the resistor R_{Psub} depends on a configuration and/or arrangement of PTAP cells. For example, referring to FIG. **1A**, the resistance of the resistor R_{Psub} increases when the distance $2*DY$ between the adjacent elongated second TAP cells **120**, **121** increase; however, the resistance of the resistor R_{Psub} decreases when the length L or the number of fin features of the elongated second TAP cells **120**, **121** increases. By configuring and/or arranging the NTAP cells and/or PTAP cells as described herein, it is possible in at least one embodiment to improve LUP immunity of the IC device **150**.

As described herein, some other approaches for TAP cell placement suffer from some potential problems. For example, in a first approach, TAP cells are placed in a half-cell height arrangement across boundaries between P wells and N wells. Such a half-cell height arrangement faces manufacturing difficulties, especially at CD below 100 nm. In contrast, as described with respect to FIGS. **1A-1B**, the TAP cells in some embodiments are completely enclosed within respective well regions, therefore avoiding manufacturing difficulties associated with the half-cell height arrangement. For another example, in the described first approach and in a different, second approach, there are concerns with respect to mixed channel effects due to implant discontinuity between closely arranged NTAP cells

and PTAP cells. Such concerns of mixed channel effects are obviated by one or more embodiments described herein. In some embodiments, it is possible to achieve one or more effects, including, but not limited to, relaxing process constraints especially at advanced manufacturing process nodes, improving latch-up immunity, reducing of areas occupied or blocked by TAP cells, and increasing of areas where standard cells other than TAP cells are placeable. In some still further embodiments, with no process constraints at advanced manufacturing process nodes, it is possible to improve latch-up immunity and/or reduce areas occupied or blocked by TAP cells. In an example, the areas occupied or blocked by TAP cells is reduced, in at least one embodiment, to about 85% of that observed in other approaches, without sacrificing LUP immunity.

FIG. **2** is a schematic view of an IC layout diagram **200**, in accordance with some embodiments. The IC layout diagram **200** comprises a plurality of portions **201**, **201**, . . . **20n** which are arranged at regular intervals in the X direction and Y direction. TAP cells are placed in each of the portions **201**, **201**, . . . **20n** in similar manner. For example, in each of the portions **201**, **201**, . . . **20n**, TAP cells are placed as described with respect to FIG. **1A** in at least one embodiment. Other TAP cell placements as described with respect to FIGS. **3A**, **3B**, and **4A**, are within the scopes of various embodiments. As a result, TAP cells are placed at regular intervals and in a repeating pattern over the IC layout diagram **200**, to ensure intended LUP immunity over the IC layout diagram **200**. In some embodiments, one or more advantages or effects described with respect to FIG. **1A** are achievable in the IC layout diagram **200**.

FIG. **3A** is a schematic view of an IC layout diagram **300A**, in accordance with some embodiments. In at least one embodiment, the IC layout diagram **300A** corresponds to any of the portions **201**, **201**, . . . **20n** in FIG. **2**. Similarly to the IC layout diagram **100**, the IC layout diagram **300A** comprises a plurality of first TAP cells representatively indicated at **310**, and a plurality of second TAP cells **320**, **321**. The first TAP cells **310** correspond to the first TAP cells **110-117** of the IC layout diagram **100**, but are arranged in more than two columns, e.g., four columns **318**, **319**, **328**, **329**. The second TAP cells **320**, **321** correspond to the second TAP cells **120**, **121**, and extend in the X direction across the four columns **318**, **319**, **328**, **329** of the first TAP cells **310**. In some embodiments, configurations, modifications, advantages or effects described with respect to FIG. **1A** are achievable in the IC layout diagram **300A**.

FIG. **3B** is a schematic view of an IC layout diagram **300B**, in accordance with some embodiments. In at least one embodiment, the IC layout diagram **300B** corresponds to any of the portions **201**, **201**, . . . **20n** in FIG. **2**. Similarly to the IC layout diagram **300A**, the IC layout diagram **300B** comprises a plurality of first TAP cells **310**, which are arranged in four columns **318**, **319**, **328**, **329**. However, instead of each of the continuous second TAP cells **320**, **321** in the IC layout diagram **300A**, the IC layout diagram **300B** includes a series **330**, **331** of discrete second TAP cells representatively indicated at **332**, **333**, **334**, **335**. The second TAP cells **332**, **333**, **334**, **335** are arranged in a line along the X direction, at an interval d . The length L of each of the series **330**, **331** for determining the number of fin features in the series is a total of length m of each second TAP cells **332**, **333**, **334**, **335** in the series. In each of the series, e.g., series **331**, a first end TAP cell **332** at a first end of the series **331** overlaps, in the Y direction, the first TAP cells in one of the columns, e.g., column **318**. A second end TAP cell **335** at a second end of the series **331** overlaps, in the Y direction, first

TAP cells in another column **329**. The series **331** further comprises at least one middle TAP cell **334** between the first and second ends of the series **331**, and overlapping no first TAP cell among the first TAP cells in the columns **318**, **319**, **328**, **329**. In some embodiments, configurations, modifications, advantages or effects described with respect to FIG. **1A** are achievable in the IC layout diagram **300B**.

FIG. **4A** is a schematic view of an IC layout diagram **400**, in accordance with some embodiments. The IC layout diagram **400** in FIG. **4A** is similar to the IC layout diagram **100** in FIG. **1A**, with the exception that P-type regions, wells or TAP cells in the IC layout diagram **400** correspond to N-type regions, wells or TAP cells in the IC layout diagram **100**, and vice versa. An element in FIG. **4A** is denoted by the same reference numeral as a corresponding element in FIG. **1A**, but with the prime symbol added to FIG. **4A**. For example, a well region **130'** in FIG. **4A** corresponds to the well region **130** in FIG. **1A**. In at least one embodiment, an IC device corresponding to the IC layout diagram **400** is formed on an N-type substrate, as described with respect to FIG. **4B**.

FIG. **4B** is a schematic cross-sectional view of an IC device **150'**, in accordance with some embodiments. The IC device **150'** corresponds to a portion of the IC layout diagram **400** which is indicated by arrow $Y' \rightarrow Y1'$ in FIG. **4A**. The cross-sectional view in FIG. **4B** is also combined with a schematic circuit diagram of the IC device **150'**. The IC device **150'** in FIG. **4B** is similar to the IC device **150** in FIG. **1C**, with the exception that P-type substrate, regions, wells, TAP cells, in the IC device **150'** correspond to N-type substrate, regions, wells, TAP cells in the IC device **150**, and vice versa. An element in FIG. **4B** is denoted by the same reference numeral as a corresponding element in FIG. **1C**, but with the prime symbol added to FIG. **4B**. For example, a substrate **151'** in FIG. **4B** corresponds to the substrate **151** in FIG. **1C**. Further, NMOS, PMOS, $Q1'(NPN)$, $Q2'(PNP)$, R_{PW} and R_{Nsub} in FIG. **4B** correspond to PMOS, NMOS, $Q1(NPN)$, $Q2(NPN)$, R_{NW} and R_{Psub} in FIG. **1C**, respectively.

In some embodiments, configurations, operations, modifications, advantages or effects described with respect to the IC layout diagram **100** in FIG. **1A** and/or the IC device **150** in FIG. **1C** are achievable in the IC layout diagram **400** in FIG. **4A** and/or the IC device **150'** in FIG. **4B**. Some embodiments include IC layout diagrams similar to the IC layout diagram **300A** or **300B**, but with similar changes from P-type substrate, wells, regions or TAP cells to N-type substrate, wells, regions or TAP cells, respectively, and vice versa.

FIG. **5** is a flow chart of a method **500** for TAP cell placement in an IC layout diagram, in accordance with some embodiments. In at least one embodiment, the method **500** is performed in whole or in part by a processor as described herein, to generate an IC layout diagram corresponding to at least one of the IC layout diagrams **100**, **200**, **300A**, **300B** and **400**.

At operation **505**, a plurality of first TAP cells is placed in an IC layout diagram such that the first TAP cells are arranged in two adjacent columns. For example, as described with respect to FIG. **1A**, a plurality of first TAP cells **110-117** is placed in the IC layout diagram **100** such that the first TAP cells **110-117** are arranged in two adjacent columns **118**, **119**. The two adjacent columns **118**, **119** are adjacent each other in a first direction, e.g., the X direction, and extend in a second direction, e.g., the Y direction,

transverse to the first direction. The first TAP cells **110-117** are of a first semiconductor type, e.g., N-type as in FIG. **1A**, or P-type as in FIG. **4A**.

At operation **515**, two second TAP cells of a second semiconductor type different from the first semiconductor type are placed in the IC layout diagram. Each of the two second TAP cells extends continuously between the two adjacent columns of the first TAP cells over a second length greater than a first length of each of the first TAP cells. For example, as described with respect to FIG. **1A**, two second TAP cells **120**, **121** are placed in the IC layout diagram **100**. Each of the two second TAP cells **120**, **121** extends continuously between the two adjacent columns **118**, **119** of the first TAP cells **110-117** over a length L greater than a length L' of each of the first TAP cells **110-117**. The two adjacent columns **118**, **119** of the first TAP cells **110-117** are located between the two second TAP cells **120**, **121** in the Y direction. The second TAP cells **120**, **121** are of a second semiconductor type, e.g., P-type as in FIG. **1A**, or N-type as in FIG. **4A**.

In at least one embodiment, operations **505** and **515** occur concurrently, e.g., in a place and route operation of an IC manufacturing flow. In one or more embodiments, the first TAP cells and/or the second TAP cells are standard cells stored in and read from one or more cell libraries. In some embodiments, operations **505** and **515** are performed to place TAP cells at regular intervals and in a repeating pattern over the IC layout diagram, as described with respect to FIG. **2**.

The described methods include example operations, but they are not necessarily required to be performed in the order shown. Operations may be added, replaced, changed order, and/or eliminated as appropriate, in accordance with the spirit and scope of embodiments of the disclosure. Embodiments that combine different features and/or different embodiments are within the scope of the disclosure and will be apparent to those of ordinary skill in the art after reviewing this disclosure.

FIG. **6** is a perspective view of an example circuit element **600** having a fin feature, in accordance with some embodiments. In the example configuration in FIG. **6**, circuit element **600** is a fin field-effect transistor (FINFET). FINFET **600** comprises a substrate **602**, at least one fin feature (or fin) **604** extending in a Z direction from substrate **602**, a gate dielectric **606** along surfaces of fin **604**, and a gate electrode **608** over gate dielectric **606**. A source region **610** and a drain region **612** are disposed over substrate **602** on opposite sides of fin **604**. Fin **604**, source region **610** and drain region **612** belong to an active region (or OD region) which corresponds, in one or more embodiments, to any active region described with respect to FIGS. **1A-4B**. In at least one embodiment, gate electrode **608** corresponds to any gate region described with respect to FIGS. **1A-4B**. The described configuration of a fin feature in an active region is an example. Other configurations are within the scopes of various embodiments.

In some embodiments, some or all of the methods discussed above are performed by an IC layout diagram generation system. In some embodiments, an IC layout diagram generation system is usable as part of a design house of an IC manufacturing system discussed below.

FIG. **7** is a block diagram of EDA system **700** in accordance with some embodiments.

In some embodiments, EDA system **700** includes an automated placement and routing (APR) system. Methods described herein of designing layout diagrams and representing wire routing arrangements, in accordance with one

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or more embodiments, are implementable, for example, using EDA system 700, in accordance with some embodiments.

In some embodiments, EDA system 700 is a general purpose computing device including a hardware processor 702 and a non-transitory, computer-readable storage medium 704. Storage medium 704, amongst other things, is encoded with, i.e., stores, computer program code 706, i.e., a set of executable instructions. Execution of instructions 706 by hardware processor 702 represents (at least in part) an EDA tool which implements a portion or all of, e.g., the methods described herein in accordance with one or more embodiments (hereinafter, the noted processes and/or methods).

Processor 702 is electrically coupled to computer-readable storage medium 704 via a bus 708. Processor 702 is also electrically coupled to an I/O interface 710 by bus 708. A network interface 712 is also electrically connected to processor 702 via bus 708. Network interface 712 is connected to a network 714, so that processor 702 and computer-readable storage medium 704 are capable of connecting to external elements via network 714. Processor 702 is configured to execute computer program code 706 encoded in computer-readable storage medium 704 in order to cause EDA system 700 to perform a portion or all of the noted processes and/or methods. In one or more embodiments, processor 702 is a central processing unit (CPU), a multi-processor, a distributed processing system, an application specific integrated circuit (ASIC), and/or a suitable processing unit.

In one or more embodiments, computer-readable storage medium 704 is an electronic, magnetic, optical, electromagnetic, infrared, and/or a semiconductor system (or apparatus or device). For example, computer-readable storage medium 704 includes a semiconductor or solid-state memory, a magnetic tape, a removable computer diskette, a random access memory (RAM), a read-only memory (ROM), a rigid magnetic disk, and/or an optical disk. In one or more embodiments using optical disks, computer-readable storage medium 704 includes a compact disk-read only memory (CD-ROM), a compact disk-read/write (CD-R/W), and/or a digital video disc (DVD).

In one or more embodiments, storage medium 704 stores computer program code 706 configured to cause EDA system 700 (where such execution represents (at least in part) the EDA tool) to perform a portion or all of the noted processes and/or methods. In one or more embodiments, storage medium 704 also stores information which facilitates performing a portion or all of the noted processes and/or methods. In one or more embodiments, storage medium 704 stores a library 707 of standard cells, including HPC cells as disclosed herein.

EDA system 700 includes I/O interface 710. I/O interface 710 is coupled to external circuitry. In one or more embodiments, I/O interface 710 includes a keyboard, keypad, mouse, trackball, trackpad, touchscreen, and/or cursor direction keys for communicating information and commands to processor 702.

EDA system 700 also includes network interface 712 coupled to processor 702. Network interface 712 allows EDA system 700 to communicate with network 714, to which one or more other computer systems are connected. Network interface 712 includes wireless network interfaces such as BLUETOOTH, WIFI, WIMAX, GPRS, or WCDMA; or wired network interfaces such as ETHERNET, USB, or IEEE-1364. In one or more embodiments, a portion

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or all of noted processes and/or methods, is implemented in two or more EDA systems 700.

EDA system 700 is configured to receive information through I/O interface 710. The information received through I/O interface 710 includes one or more of instructions, data, design rules, libraries of standard cells, and/or other parameters for processing by processor 702. The information is transferred to processor 702 via bus 708. EDA system 700 is configured to receive information related to a UI through I/O interface 710. The information is stored in computer-readable medium 704 as user interface (UI) 742.

In some embodiments, a portion or all of the noted processes and/or methods is implemented as a standalone software application for execution by a processor. In some embodiments, a portion or all of the noted processes and/or methods is implemented as a software application that is a part of an additional software application. In some embodiments, a portion or all of the noted processes and/or methods is implemented as a plug-in to a software application. In some embodiments, at least one of the noted processes and/or methods is implemented as a software application that is a portion of an EDA tool. In some embodiments, a portion or all of the noted processes and/or methods is implemented as a software application that is used by EDA system 700. In some embodiments, a layout diagram which includes standard cells is generated using a tool such as VIRTUOSO® available from CADENCE DESIGN SYSTEMS, Inc., or another suitable layout generating tool.

In some embodiments, the processes are realized as functions of a program stored in a non-transitory computer readable recording medium. Examples of a non-transitory computer readable recording medium include, but are not limited to, external/removable and/or internal/built-in storage or memory unit, e.g., one or more of an optical disk, such as a DVD, a magnetic disk, such as a hard disk, a semiconductor memory, such as a ROM, a RAM, a memory card, and the like.

FIG. 8 is a block diagram of an integrated circuit (IC) manufacturing system 800, and an IC manufacturing flow associated therewith, in accordance with some embodiments. In some embodiments, based on a layout diagram, at least one of (A) one or more semiconductor masks or (B) at least one component in a layer of a semiconductor integrated circuit is fabricated using manufacturing system 800.

In FIG. 8, IC manufacturing system 800 includes entities, such as a design house 820, a mask house 830, and an IC manufacturer/fabricator ("fab") 850, that interact with one another in the design, development, and manufacturing cycles and/or services related to manufacturing an IC device 860. The entities in system 800 are connected by a communications network. In some embodiments, the communications network is a single network. In some embodiments, the communications network is a variety of different networks, such as an intranet and the Internet. The communications network includes wired and/or wireless communication channels. Each entity interacts with one or more of the other entities and provides services to and/or receives services from one or more of the other entities. In some embodiments, two or more of design house 820, mask house 830, and IC fab 850 is owned by a single larger company. In some embodiments, two or more of design house 820, mask house 830, and IC fab 850 coexist in a common facility and use common resources.

Design house (or design team) 820 generates an IC design layout diagram 822. IC design layout diagram 822 includes various geometrical patterns designed for IC device 860. The geometrical patterns correspond to patterns of metal,

oxide, or semiconductor layers that make up the various components of IC device **860** to be fabricated. The various layers combine to form various IC features. For example, a portion of IC design layout diagram **822** includes various IC features, such as an active region, gate electrode, source and drain, metal lines or vias of an interlayer interconnection, and openings for bonding pads, to be formed in a semiconductor substrate (such as a silicon wafer) and various material layers disposed on the semiconductor substrate. Design house **820** implements a proper design procedure to form IC design layout diagram **822**. The design procedure includes one or more of logic design, physical design or place and route. IC design layout diagram **822** is presented in one or more data files having information of the geometrical patterns. For example, IC design layout diagram **822** is expressed in a GDSII file format or DFII file format.

Mask house **830** includes data preparation **832** and mask fabrication **844**. Mask house **830** uses IC design layout diagram **822** to manufacture one or more masks **845** to be used for fabricating the various layers of IC device **860** according to IC design layout diagram **822**. Mask house **830** performs mask data preparation **832**, where IC design layout diagram **822** is translated into a representative data file ("RDF"). Mask data preparation **832** provides the RDF to mask fabrication **844**. Mask fabrication **844** includes a mask writer. A mask writer converts the RDF to an image on a substrate, such as a mask (reticle) **845** or a semiconductor wafer **853**. The design layout diagram **822** is manipulated by mask data preparation **832** to comply with particular characteristics of the mask writer and/or requirements of IC fab **850**. In FIG. 8, mask data preparation **832** and mask fabrication **844** are illustrated as separate elements. In some embodiments, mask data preparation **832** and mask fabrication **844** can be collectively referred to as mask data preparation.

In some embodiments, mask data preparation **832** includes optical proximity correction (OPC) which uses lithography enhancement techniques to compensate for image errors, such as those that can arise from diffraction, interference, other process effects and the like. OPC adjusts IC design layout diagram **822**. In some embodiments, mask data preparation **832** includes further resolution enhancement techniques (RET), such as off-axis illumination, sub-resolution assist features, phase-shifting masks, other suitable techniques, and the like or combinations thereof. In some embodiments, inverse lithography technology (ILT) is also used, which treats OPC as an inverse imaging problem.

In some embodiments, mask data preparation **832** includes a mask rule checker (MRC) that checks the IC design layout diagram **822** that has undergone processes in OPC with a set of mask creation rules which contain certain geometric and/or connectivity restrictions to ensure sufficient margins, to account for variability in semiconductor manufacturing processes, and the like. In some embodiments, the MRC modifies the IC design layout diagram **822** to compensate for limitations during mask fabrication **844**, which may undo part of the modifications performed by OPC in order to meet mask creation rules.

In some embodiments, mask data preparation **832** includes lithography process checking (LPC) that simulates processing that will be implemented by IC fab **850** to fabricate IC device **860**. LPC simulates this processing based on IC design layout diagram **822** to create a simulated manufactured device, such as IC device **860**. The processing parameters in LPC simulation can include parameters associated with various processes of the IC manufacturing cycle, parameters associated with tools used for manufacturing the

IC, and/or other aspects of the manufacturing process. LPC takes into account various factors, such as aerial image contrast, depth of focus ("DOF"), mask error enhancement factor ("MEEF"), other suitable factors, and the like or combinations thereof. In some embodiments, after a simulated manufactured device has been created by LPC, if the simulated device is not close enough in shape to satisfy design rules, OPC and/or MRC are repeated to further refine IC design layout diagram **822**.

It should be understood that the above description of mask data preparation **832** has been simplified for the purposes of clarity. In some embodiments, data preparation **832** includes additional features such as a logic operation (LOP) to modify the IC design layout diagram **822** according to manufacturing rules. Additionally, the processes applied to IC design layout diagram **822** during data preparation **832** may be executed in a variety of different orders.

After mask data preparation **832** and during mask fabrication **844**, a mask **845** or a group of masks **845** are fabricated based on the modified IC design layout diagram **822**. In some embodiments, mask fabrication **844** includes performing one or more lithographic exposures based on IC design layout diagram **822**. In some embodiments, an electron-beam (e-beam) or a mechanism of multiple e-beams is used to form a pattern on a mask (photomask or reticle) **845** based on the modified IC design layout diagram **822**. Mask **845** can be formed in various technologies. In some embodiments, mask **845** is formed using binary technology. In some embodiments, a mask pattern includes opaque regions and transparent regions. A radiation beam, such as an ultraviolet (UV) beam, used to expose the image sensitive material layer (e.g., photoresist) which has been coated on a wafer, is blocked by the opaque region and transmits through the transparent regions. In one example, a binary mask version of mask **845** includes a transparent substrate (e.g., fused quartz) and an opaque material (e.g., chromium) coated in the opaque regions of the binary mask. In another example, mask **845** is formed using a phase shift technology. In a phase shift mask (PSM) version of mask **845**, various features in the pattern formed on the phase shift mask are configured to have proper phase difference to enhance the resolution and imaging quality. In various examples, the phase shift mask can be attenuated PSM or alternating PSM. The mask(s) generated by mask fabrication **844** is used in a variety of processes. For example, such a mask(s) is used in an ion implantation process to form various doped regions in semiconductor wafer **853**, in an etching process to form various etching regions in semiconductor wafer **853**, and/or in other suitable processes.

IC fab **850** includes wafer fabrication **852**. IC fab **850** is an IC fabrication business that includes one or more manufacturing facilities for the fabrication of a variety of different IC products. In some embodiments, IC Fab **850** is a semiconductor foundry. For example, there may be a manufacturing facility for the front end fabrication of a plurality of IC products (front-end-of-line (FEOL) fabrication), while a second manufacturing facility may provide the back end fabrication for the interconnection and packaging of the IC products (back-end-of-line (BEOL) fabrication), and a third manufacturing facility may provide other services for the foundry business.

IC fab **850** uses mask(s) **845** fabricated by mask house **830** to fabricate IC device **860**. Thus, IC fab **850** at least indirectly uses IC design layout diagram **822** to fabricate IC device **860**. In some embodiments, semiconductor wafer **853** is fabricated by IC fab **850** using mask(s) **845** to form IC device **860**. In some embodiments, the IC fabrication

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includes performing one or more lithographic exposures based at least indirectly on IC design layout diagram **822**. Semiconductor wafer **853** includes a silicon substrate or other proper substrate having material layers formed thereon. Semiconductor wafer **853** further includes one or more of various doped regions, dielectric features, multi-level interconnects, and the like (formed at subsequent manufacturing steps).

Details regarding an integrated circuit (IC) manufacturing system (e.g., system **800** of FIG. **8**), and an IC manufacturing flow associated therewith are found, e.g., in U.S. Pat. No. 9,256,709, granted Feb. 9, 2016, U.S. Pre-Grant Publication No. 20150278429, published Oct. 1, 2015, U.S. Pre-Grant Publication No. 20140040838, published Feb. 6, 2014, and U.S. Pat. No. 7,260,442, granted Aug. 21, 2007, the entireties of each of which are hereby incorporated by reference.

In some embodiments, an integrated circuit (IC) device comprises a plurality of first doped regions of a first semiconductor type over at least one first well region of the first semiconductor type, and a second doped region of a second semiconductor type over a second well region of the second semiconductor type, the second semiconductor type different from the first semiconductor type. The plurality of first doped regions is arranged along a first direction. Each of the plurality of first doped regions has a first length in the first direction. The second doped region extends in the first direction between at least two first doped regions among the plurality of first doped regions over a second length greater than the first length.

In some embodiments, a method of manufacturing an integrated circuit (IC) device comprises forming, over a substrate, a plurality of first well regions of a first semiconductor type and forming, over the substrate, two second well regions of a second semiconductor type different from the first semiconductor type. The method further comprises forming a plurality of first doped regions of the first semiconductor type over the plurality of first well regions, and forming two second doped regions of the second semiconductor type correspondingly over the two second well regions. The plurality of first well regions and the two second well regions are elongated along a first direction. The plurality of first well regions are arranged, in a second direction transverse to the first direction, between the two second well regions. Each of the plurality of first doped regions has a first length in the first direction. Each of the two second doped regions has, in the first direction, a second length greater than the first length.

In some embodiments, a method of manufacturing an integrated circuit (IC) device comprises forming, over a substrate, a first well region of a first semiconductor type, and forming, over the substrate, a second well region of a second semiconductor type different from the first semiconductor type. The first well region and the second well region are elongated along a first direction. The method further comprises forming a plurality of first doped regions of the first semiconductor type over the first well region and at an interval along the first direction, and forming a second doped region of the second semiconductor type over the second well region. The second doped region is elongated in the first direction and overlaps, in a second direction transverse to the first direction, multiple first doped regions among the plurality of first doped regions.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other

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processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. An integrated circuit (IC) device, comprising:

a plurality of first doped regions of a first semiconductor type over at least one first well region of the first semiconductor type; and

a second doped region of a second semiconductor type over a second well region of the second semiconductor type, the second semiconductor type different from the first semiconductor type,

wherein

the plurality of first doped regions is arranged along a first direction,

each of the plurality of first doped regions has a first length in the first direction,

the second doped region extends in the first direction between at least two first doped regions among the plurality of first doped regions over a second length greater than the first length,

the plurality of first doped regions is configured to receive a first power supply voltage,

the second doped region is configured to receive a second power supply voltage different from the first power supply voltage,

the plurality of first doped regions is arranged in at least two columns which are adjacent each other in the first direction and elongated in a second direction transverse to the first direction, and

each column of the at least two columns includes multiple first doped regions among the plurality of first doped regions, wherein two immediately adjacent first doped regions among the multiple first doped regions in the column are separated from each other, in the second direction, by a region containing dopants of the second semiconductor type.

2. The IC device of claim 1, further comprising:

a substrate on which the plurality of first doped regions and the second doped region are arranged,

wherein the substrate is of the second semiconductor type.

3. The IC device of claim 1, further comprising:

another second doped region of the second semiconductor type over another second well region of the second semiconductor type, the another second doped region extending in the first direction between the at least two first doped regions over the second length greater than the first length,

the another second doped region is configured to receive the second power supply voltage, and

the plurality of first doped regions is arranged in the second direction between the second doped region and the another second doped region.

4. The IC device of claim 3, further comprising:

a plurality of first well regions of the first semiconductor type and including the at least one first well region, each of the plurality of first doped regions in a corresponding one of the plurality of first well regions;

a plurality of second well regions of the second semiconductor type and including the second well region and the another second well region, the plurality of first well regions and the plurality of second well regions

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extending in the first direction and arranged alternat-
ingly in the second direction; and
fin features extending in the first direction,
wherein

$$F2/F1 \geq (DY/DX) * (L/A),$$

where

F1 is a number of fin features in each of the plurality of
first doped regions,

F2 is a number of fin features in each of the second doped
region and the another second doped region,

DX is a half of a first distance in the first direction
between facing sides of the at least two first doped
regions,

DY is a half of a second distance in the second direction
between facing sides of the second doped region and
the another second doped region,

L is the second length of the second doped region or the
another second doped region in the first direction, and

A is a sum of a height, in the second direction, of a first
well region among the plurality of first well regions and
a height, in the second direction, of an adjacent second
well region among the plurality of second well regions.

5. The IC device of claim 4, satisfying at least one of

DY is from 0.5 μm to 1000 μm ,

DX is from 0.05 μm to 100 μm ,

L is from 0.1 μm to 5000 μm , or

A is from 0.025 μm to 0.300 μm .

6. The IC device of claim 4, further comprising:
gate regions extending in the second direction,
wherein each of F1 and F2 is defined by

$$(L_{OD}/CPP) * W * F_n,$$

where

L_{OD} is the first length of a corresponding first doped
region, or the second length of the second doped region
or the another second doped region,

CPP is a pitch between adjacent gate regions in the first
direction,

W is the height of the corresponding first doped region,
the second doped region, or the another second doped
region in the second direction, and

F_n is a number of fin features per unit height in the second
direction.

7. The IC device of claim 1, further comprising:

a plurality of first well regions of the first semiconductor
type and including the at least one first well region,
each of the plurality of first doped regions in a corre-
sponding one of the plurality of first well regions;

a plurality of second well regions of the second semicon-
ductor type and including the second well region; and

a substrate on which the plurality of first well regions, the
plurality of second well regions, the plurality of first
doped regions and the second doped region are
arranged,

wherein

the first semiconductor type is N-type, the second
semiconductor type is P-type, and the substrate is a
P-type substrate, or

the first semiconductor type is P-type, the second
semiconductor type is N-type, and the substrate is an
N-type substrate.

8. The IC device of claim 1, wherein

the second doped region overlaps, in the second direction,
at least one first doped region among the plurality of
first doped regions.

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9. The IC device of claim 1, wherein

the second doped region overlaps, in the second direction,
at least one column of the at least two columns in which
the plurality of first doped regions is arranged.

10. The IC device of claim 1, wherein

the second doped region extends continuously in the first
direction between first and second ends thereof,

the first end of the second doped region overlaps, in the
second direction, one of the at least two first doped
regions, and

the second end of the second doped region overlaps, in the
second direction, another one of the at least two first
doped regions.

11. The IC device of claim 1, wherein

the second doped region includes a series of second doped
regions arranged along the first direction,

two immediately adjacent second doped regions in the
series of second doped regions are separated from each
other, in the first direction, by a section of the second
well region which contains dopants of the second
semiconductor type at a concentration lower than the
second doped regions in the series, and

the series of second doped regions includes:

a first end doped region at a first end of the series of
second doped regions and overlapping, in the second
direction, one of the at least two first doped regions,

a second end doped region at a second end of the series
of second doped regions and overlapping, in the
second direction, another one of the at least two first
doped regions, and

a middle doped region between the first and second
ends of the series of second doped regions and
overlapping, in the second direction, no first doped
region among the plurality of first doped regions.

12. The IC device of claim 1, wherein

the at least two columns comprise more than two col-
umns, and

the plurality of first doped regions in the more than two
columns overlap, in the second direction, the second
doped region.

13. An integrated circuit (IC) device, comprising:

a substrate;

over the substrate, a plurality of first well regions of a first
semiconductor type;

over the substrate, a plurality of second well regions of a
second semiconductor type different from the first
semiconductor type, wherein

the plurality of first well regions and the plurality of
second well regions are elongated along a first direc-
tion,

the plurality of first well regions and the plurality of
second well regions are alternately arranged along
a second direction transverse to the first direction,
and

two immediately adjacent first well regions among the
plurality of first well regions are separated from each
other, in the second direction, by a second well region
among the plurality of second well regions;

a plurality of first doped regions of the first semiconductor
type over the plurality of first well regions; and

two second doped regions of the second semiconductor
type correspondingly over two end second well regions
among the plurality of second well regions,

wherein

the plurality of first well regions and one or more further
second well regions among the plurality of second well
regions are arranged, in the second direction, between
the two end second well regions,

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each of the plurality of first doped regions has a first length in the first direction,

each of the two second doped regions has, in the first direction, a second length greater than the first length.

14. The IC device of claim 13, wherein
no other doped regions of the second semiconductor type are arranged, in the second direction, between the two second doped regions.

15. The IC device of claim 13, wherein
each of the two second doped regions has
a first end overlapping, in the second direction, some first doped regions among the plurality of first doped regions, and
a second end overlapping, in the second direction, some further first doped regions among the plurality of first doped regions.

16. The IC device of claim 13, wherein
the plurality of first doped regions is arranged in two adjacent columns, which are adjacent each other in the first direction, and
each column of the two adjacent columns includes multiple first doped regions among the plurality of first doped regions, the multiple first doped regions arranged in the column along the second direction.

17. The IC device of claim 16, further comprising:
fin features extending in the first direction, wherein

$$F2/F1 \geq (DY/DX) * (L/A),$$

where

F1 is a number of fin features in each of the plurality of first doped regions,

F2 is a number of fin features in each of the two second doped regions,

DX is a half of a first distance in the first direction between facing sides of the two adjacent columns,

DY is a half of a second distance in the second direction between facing sides of the two second doped regions,

L is the second length of each of the two second doped regions in the first direction, and

A is a sum of a height, in the second direction, of a first well region among the plurality of first well regions and a height, in the second direction, of an adjacent second well region among the plurality of second well regions.

18. The IC device of claim 13, wherein
the plurality of first doped regions and the two second doped regions are arranged at regular intervals in the first direction and the second direction over an area of the IC device.

19. An integrated circuit (IC) device, comprising:
a substrate;
over the substrate, a first well region of a first semiconductor type;

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over the substrate, a second well region of a second semiconductor type different from the first semiconductor type, wherein the first well region and the second well region are elongated along a first direction;

a plurality of first doped regions of the first semiconductor type over the first well region, wherein the plurality of first doped regions is discontinuous from each other and arranged at a first interval along the first direction; and

a plurality of second doped regions of the second semiconductor type over the second well region, wherein the plurality of second doped regions is discontinuous from each other and arranged, along the first direction, at a second interval smaller than the first interval,

wherein

the first interval along the first direction between two immediately adjacent first doped regions among the plurality of first doped regions is a section of the first well region which contains dopants of the first semiconductor type at a concentration lower than the plurality of first doped regions, and the section of the first well region overlaps, in a second direction transverse to the first direction, multiple second doped regions among the plurality of second doped regions, and

the second interval along the first direction between two immediately adjacent second doped regions among the plurality of second doped regions is a section of the second well region which contains dopants of the second semiconductor type at a concentration lower than the plurality of second doped regions.

20. The IC device of claim 19, further comprising:
over the substrate, a further second well region of the second semiconductor type; and

a plurality of further second doped regions of the second semiconductor type over the further second well region,

wherein

the plurality of further second doped regions is discontinuous from each other and arranged at the second interval along the first direction,

the second interval between two immediately adjacent further second doped regions among the plurality of further second doped regions is a section of the further second well region which contains dopants of the second semiconductor type at a concentration lower than the plurality of further second doped regions, and one second doped region among the plurality of second doped regions and a further second doped region among the plurality of further second doped regions are adjacent each other in the second direction, and sandwich therebetween multiple first doped regions of the plurality of first doped regions.

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